Python Program – Bayesian Network

```
# Bayesian Network
# Rain -> Wet Grass
# Sprinkler -> Wet Grass

# Prior probabilities
P_Rain = 0.2
P_Sprinkler = 0.3

# Conditional Probability Table
# P(Wet Grass | Rain, Sprinkler)

cpt = {
   (True, True): 0.99,
   (True, False): 0.90,
   (False, True): 0.80,
   (False, False): 0.01
}

# Calculate probability of Wet Grass
P_WetGrass = 0

for rain in [True, False]:
   for sprinkler in [True, False]:

      p_rain = P_Rain if rain else 1 - P_Rain

      p_sprinkler = (
         P_Sprinkler if sprinkler
```

```
            else 1 - P_Sprinkler
        )

        p_wet = cpt[(rain, sprinkler)]

        probability = (
            p_rain * p_sprinkler * p_wet
        )

        P_WetGrass += probability

print("Probability of Wet Grass:",
      round(P_WetGrass, 4))
```

Output

Probability of Wet Grass: 0.2781

Python Program – Bayesian Network using Conditional Probability

```
P_Rain = 0.2
P_Sprinkler = 0.3

cpt = {
    (True, True): 0.99,
    (True, False): 0.90,
    (False, True): 0.80,
    (False, False): 0.01
}

P_WetGrass = 0
```

```
P_Rain_And_WetGrass = 0

for rain in [True, False]:
    for sprinkler in [True, False]:

        pr = P_Rain if rain else 1 - P_Rain
        ps = P_Sprinkler if sprinkler else 1 - P_Sprinkler

        pw = cpt[(rain, sprinkler)]

        joint = pr * ps

        P_WetGrass += joint * pw

        if rain:
            P_Rain_And_WetGrass += joint * pw

result = P_Rain_And_WetGrass / P_WetGrass

print("P(Wet Grass):", round(P_WetGrass, 4))
print("P(Rain and Wet Grass):",
      round(P_Rain_And_WetGrass, 4))
print("P(Rain | Wet Grass):", round(result, 4))
```

Output

P(Wet Grass): 0.2781

P(Rain and Wet Grass): 0.18

P(Rain | Wet Grass): 0.6472